Attendance Summary Dialog Misalignment Fix & UI Audit

Verification Report for Faculty & Class Adviser Confirmation Dialog

Component: Faculty / Class Adviser Daily Summary Dialog	**Audit Date:** 2026-10-19
Target Device: Samsung Galaxy S24 (SM-S921E)	**Platform:** Flutter Mobile (Android/iOS/Web)
Affected Files: faculty_class_advisor_entry_screen.dart, faculty_class_advisor_screen.dart	**Status:** VERIFIED & RESOLVED

1. Bug Description & Root Cause Analysis

In the Faculty / Class Adviser portal, upon saving attendance or opening the daily summary action, a confirmation dialog titled **'Attendance Saved — Daily Summary'** is presented to the user. A screenshot submitted by QA revealed severe visual layout defects on mobile viewports:

- **Horizontal Metric Overlap:** Attendance count labels severely collided horizontally. Labels including *'Full Day Present'*, *'Half-Day Present'*, *'Full Day Absent'*, *'On Duty (OD)'*, *'Partial / Mixed'*, and *'Not Entered'* collided into contiguous strings such as *'Full Day PresentHalf-Day PresentFull Day Ab'*.
- **Inadequate Column Sizing:** The metrics were laid out in two Row(mainAxisAlignment: MainAxisAlignment.spaceAround) containers with 3 unconstrained columns per row inside a dialog with ~280px-300px available width. Each label requires ~100-120px to display cleanly, causing catastrophic text collision.
- **Duplicated Text in Lunch Break Card:** The backend response model provided lunchInterval = '12:50 PM - 01:40 PM (Not Counted)'. The UI layer previously appended ' • Not counted', producing redundant text: *'Lunch Break: 12:50 PM - 01:40 PM (Not Counted) • Not counted'*.
- **Dialog Padding & Scroll Constraints:** The default AlertDialog horizontal padding of 40px severely squeezed content on narrow mobile screens, and unbounded content lacked height constraints.

2. Architectural & Engineering Solution

To provide a permanent, robust fix across all screen sizes and orientations without resorting to crude font shrinkage, a dedicated domain widget was engineered in mobile/lib/widgets/attendance/daily_attendance_summary_metrics.dart:

Component	Engineering Implementation	Result & Benefit
DailyAttendanceSummaryMetrics	Responsive layout evaluating viewport width (MediaQuery.sizeOf(context).width). Renders a **2-column by 3-row grid** on mobile phones (<460px) and a **3-column by 2-row grid** on tablets/desktop (≥460px). Uses Expanded tiles with minHeight: 70.	Ample 140px-170px width per card on mobile phones. Zero horizontal collision. Prominent 22pt bold numeric count at top, clean multi-line wrapped caption below.

DailyAttendanceLunchBreak Card	Normalizes the lunch interval string using regex (replaceAll(RegExp(r'\s*\([Nn]ot [Cc]ounted\)'), '')) before rendering 'Lunch Break: $cleanLunch • Not counted'.	Eliminates duplicate '(Not Counted)' text entirely. Clean amber badge with restaurant icon and high contrast.
Dialog Container & Padding	Custom insetPadding: EdgeInsets.symmetric(horizontal: 18, vertical: 24), titlePadding, contentPadding, and BoxConstraints(maxHeight: height * 0.75).	Increases usable dialog width on phone screens by 44px (from 280px to 324px), guarantees zero vertical overflow on compact screens.
Unified Action Button	Full-width pill ElevatedButton with primary brand color, bold typography, and 12px vertical padding.	Meets ergonomic mobile touch target standards (≥48dp) and consistent visual hierarchy.

3. Verification, Static Analysis & Automated Testing

A rigorous verification protocol was executed across Dart static analysis, automated unit/widget testing, and integration testing:

Test Suite / Verification Gate	Scope & Test Conditions	Outcome
Flutter Static Analysis	flutter analyze lib/widgets/ lib/faculty/screens/	**PASS (0 issues found)**
Mobile 2-Col Layout Test	Renders all 6 metrics on mobile viewport (width: 300px). Verifies 3 rows of 2 columns, all 6 counts and labels present with zero overflow.	**PASS**
Tablet 3-Col Layout Test	Renders 6 metrics on tablet viewport (width: 500px). Verifies 2 rows of 3 columns, clean responsive switch.	**PASS**
Dark Mode Theme Test	Validates contrast, token colors (present, error, warning, textSecondary) under dark theme.	**PASS**
Lunch Break Deduplication Test	Tests both '12:50 PM - 01:40 PM (Not Counted)' and raw strings. Asserts deduplication works cleanly.	**PASS**
Full Dialog Integration Test	Simulates small screen (360x640), opens dialog via tap, asserts all headers, metrics, lunch card, and OK button dismiss cleanly.	**PASS**
Faculty Advisor Regression Tests	Runs faculty_class_advisor_lunch_break_removal_test.dart and unmarked_class_advisor_entry_test.dart.	**PASS (6 of 6 passed)**

4. The Six Daily Summary Metrics Specification

Full Day Present	colors.present (#16A34A)	22pt Bold, Primary Accent Tile	Wraps cleanly across 2 lines without overlap.
Half-Day Present	colors.present (#16A34A)	22pt Bold, Primary Accent Tile	Wraps cleanly across 2 lines without overlap.
Full Day Absent	colors.error (#DC2626)	22pt Bold, Destructive Alert Tile	Wraps cleanly across 2 lines without overlap.
On Duty (OD)	colors.warning (#D97706)	22pt Bold, Status Warning Tile	Single line or wrap, high legibility.
Partial / Mixed	colors.warning (#D97706)	22pt Bold, Status Warning Tile	Wraps cleanly across 2 lines without overlap.
Not Entered	colors.textMuted (#94A3B8)	22pt Bold, Neutral Tile	Single line or wrap, muted background.

QA Engineering Verification: All 6 metrics render with zero text collision, zero overflow, and clean wrapping across phones and tablets.

Sign-off: **PASSED & READY FOR RELEASE**